

Week 3: Answer Key

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```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
  filter, lag
```

```
The following objects are masked from 'package:base':
```

```
  intersect, setdiff, setequal, union
```

Exercise 1

```
Rows: 15 Columns: 5
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
dbl (5): site, experiment, length, width, height
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
[1] "1a"
```

```
# A tibble: 15 x 1
```

```
  length
```

```
  <dbl>
```

```
1    2.2
```

```
2    2.1
```

```
3    2.7
```

4	3
5	3.1
6	2.5
7	1.9
8	1.1
9	3.5
10	2.9
11	4.5
12	1.2
13	2.6
14	1.8
15	3.1

[1] "1b"

```
# A tibble: 15 x 2
  site experiment
  <dbl>         <dbl>
1     1           1
2     1           2
3     1           3
4     2           1
5     2           2
6     2           3
7     3           1
8     3           2
9     3           3
10    4           1
11    4           2
12    4           3
13    5           1
14    5           2
15    5           3
```

[1] "1c"

```
# A tibble: 15 x 6
  site experiment length width height area
  <dbl>         <dbl> <dbl> <dbl> <dbl> <dbl>
1     1           1   2.2   1.3   9.6  2.86
2     1           2   2.1   2.2   7.6  4.62
```

3	1	3	2.7	1.5	2.2	4.05
4	2	1	3	4.5	1.5	13.5
5	2	2	3.1	3.1	4	9.61
6	2	3	2.5	2.8	3	7
7	3	1	1.9	1.8	4.5	3.42
8	3	2	1.1	0.5	2.3	0.55
9	3	3	3.5	2	7.5	7
10	4	1	2.9	2.7	3.2	7.83
11	4	2	4.5	4.8	6.5	21.6
12	4	3	1.2	1.8	2.7	2.16
13	5	1	2.6	0.8	NA	2.08
14	5	2	1.8	NA	5.2	NA
15	5	3	3.1	2.2	NA	6.82

[1] "1d"

A tibble: 15 x 5

	site	experiment	length	width	height
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	3	2	1.1	0.5	2.3
2	4	3	1.2	1.8	2.7
3	5	2	1.8	NA	5.2
4	3	1	1.9	1.8	4.5
5	1	2	2.1	2.2	7.6
6	1	1	2.2	1.3	9.6
7	2	3	2.5	2.8	3
8	5	1	2.6	0.8	NA
9	1	3	2.7	1.5	2.2
10	4	1	2.9	2.7	3.2
11	2	1	3	4.5	1.5
12	2	2	3.1	3.1	4
13	5	3	3.1	2.2	NA
14	3	3	3.5	2	7.5
15	4	2	4.5	4.8	6.5

[1] "1e"

A tibble: 5 x 5

	site	experiment	length	width	height
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	1	2.2	1.3	9.6

2	1	2	2.1	2.2	7.6
3	3	3	3.5	2	7.5
4	4	2	4.5	4.8	6.5
5	5	2	1.8	NA	5.2

[1] "1f"

```
# A tibble: 2 x 5
  site experiment length width height
<dbl>      <dbl> <dbl> <dbl> <dbl>
1     1          2   2.1  2.2   7.6
2     4          2   4.5  4.8   6.5
```

[1] "1g"

```
# A tibble: 10 x 5
  site experiment length width height
<dbl>      <dbl> <dbl> <dbl> <dbl>
1     1          1   2.2  1.3   9.6
2     1          3   2.7  1.5   2.2
3     2          1    3    4.5   1.5
4     2          3   2.5  2.8    3
5     3          1   1.9  1.8   4.5
6     3          3   3.5  2    7.5
7     4          1   2.9  2.7   3.2
8     4          3   1.2  1.8   2.7
9     5          1   2.6  0.8  NA
10    5          3   3.1  2.2  NA
```

[1] "1h"

```
# A tibble: 13 x 5
  site experiment length width height
<dbl>      <dbl> <dbl> <dbl> <dbl>
1     1          1   2.2  1.3   9.6
2     1          2   2.1  2.2   7.6
3     1          3   2.7  1.5   2.2
4     2          1    3    4.5   1.5
5     2          2   3.1  3.1    4
6     2          3   2.5  2.8    3
7     3          1   1.9  1.8   4.5
```

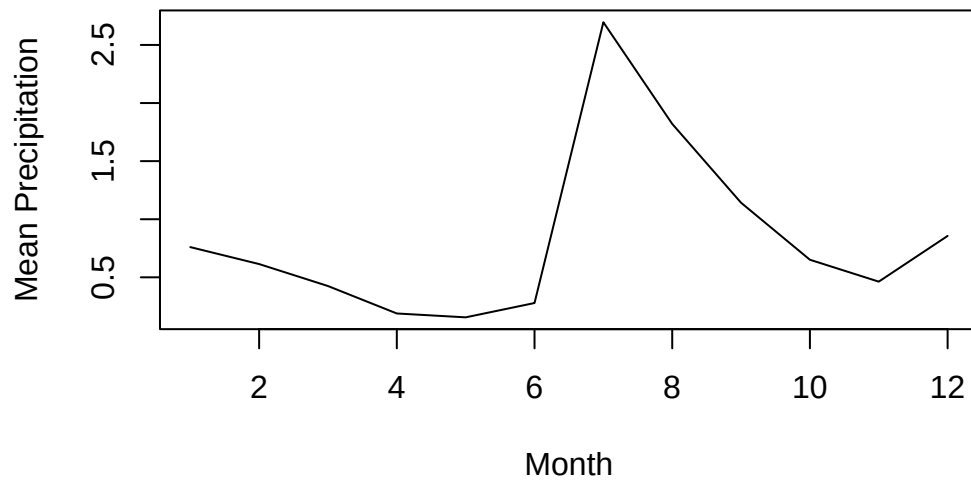
8	3	2	1.1	0.5	2.3
9	3	3	3.5	2	7.5
10	4	1	2.9	2.7	3.2
11	4	2	4.5	4.8	6.5
12	4	3	1.2	1.8	2.7
13	5	2	1.8	NA	5.2

```
[1] "1i"
```

```
# A tibble: 15 x 6
```

	site	experiment	length	width	height	volume
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	1	2.2	1.3	9.6	27.5
2	1	2	2.1	2.2	7.6	35.1
3	1	3	2.7	1.5	2.2	8.91
4	2	1	3	4.5	1.5	20.2
5	2	2	3.1	3.1	4	38.4
6	2	3	2.5	2.8	3	21
7	3	1	1.9	1.8	4.5	15.4
8	3	2	1.1	0.5	2.3	1.26
9	3	3	3.5	2	7.5	52.5
10	4	1	2.9	2.7	3.2	25.1
11	4	2	4.5	4.8	6.5	140.
12	4	3	1.2	1.8	2.7	5.83
13	5	1	2.6	0.8	NA	NA
14	5	2	1.8	NA	5.2	NA
15	5	3	3.1	2.2	NA	NA

Exercise 2



Exercise 3

Rows: 35549 Columns: 9

-- Column specification -----

Delimiter: ","

chr (2): species_id, sex

dbl (7): record_id, month, day, year, plot_id, hindfoot_length, weight

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

[1] "3a"

A tibble: 35,549 x 4

	year	month	day	species_id
	<dbl>	<dbl>	<dbl>	<chr>
1	1977	7	16	NL
2	1977	7	16	NL
3	1977	7	16	DM
4	1977	7	16	DM
5	1977	7	16	DM
6	1977	7	16	PF

```

7 1977      7    16 PE
8 1977      7    16 DM
9 1977      7    16 DM
10 1977     7    16 PF
# i 35,539 more rows

```

```
[1] "3b"
```

```

# A tibble: 32,283 x 4
  year species_id weight weight_kg
  <dbl> <chr>      <dbl>    <dbl>
1  1977 DM         40     0.04
2  1977 DM         48     0.048
3  1977 DM         29     0.029
4  1977 DM         46     0.046
5  1977 DM         36     0.036
6  1977 D0         52     0.052
7  1977 PF          8     0.008
8  1977 OX         22     0.022
9  1977 DM         35     0.035
10 1977 PF          7     0.007
# i 32,273 more rows

```

```
[1] "3c"
```

```

# A tibble: 141 x 4
  year species_id weight weight_kg
  <dbl> <chr>      <dbl>    <dbl>
1  1978 SH         89     0.089
2  1982 SH        106     0.106
3  1982 SH         52     0.052
4  1986 SH         55     0.055
5  1987 SH         77     0.077
6  1987 SH         78     0.078
7  1987 SH        104     0.104
8  1987 SH         58     0.058
9  1987 SH         52     0.052
10 1988 SH         60     0.06
# i 131 more rows

```

Exercise 4

```
[1] "4a"
```

```
# A tibble: 32,283 x 4
  year species_id weight weight_kg
  <dbl> <chr>      <dbl>    <dbl>
1  1977 DM         40     0.04
2  1977 DM         48     0.048
3  1977 DM         29     0.029
4  1977 DM         46     0.046
5  1977 DM         36     0.036
6  1977 DO         52     0.052
7  1977 PF          8     0.008
8  1977 OX         22     0.022
9  1977 DM         35     0.035
10 1977 PF          7     0.007
# i 32,273 more rows
```

```
[1] "4b"
```

```
# A tibble: 147 x 4
  year month   day species_id
  <dbl> <dbl> <dbl> <chr>
1  1977     7    17 SH
2  1978    11     4 SH
3  1982     5    21 SH
4  1982     6    29 SH
5  1983     3    14 SH
6  1983     4    16 SH
7  1986    10     4 SH
8  1987     7    26 SH
9  1987     8    26 SH
10 1987    10    24 SH
# i 137 more rows
```

Exercise 5

```
# A tibble: 14,558 x 6
  month   day year species_id weight hindfoot_length
  <dbl> <dbl> <dbl> <chr>      <dbl>      <dbl>
1     4    29 1979 dm         65         37
2     8     7 1991 dm         65         37
```

3	5	16	2002	dm	64	35
4	5	13	1984	dm	63	35
5	12	3	1995	dm	63	38
6	10	12	1980	dm	62	35
7	10	28	1995	dm	62	37
8	1	28	1996	dm	62	38
9	1	28	1996	dm	62	38
10	11	7	1999	dm	62	36
# i 14,548 more rows						